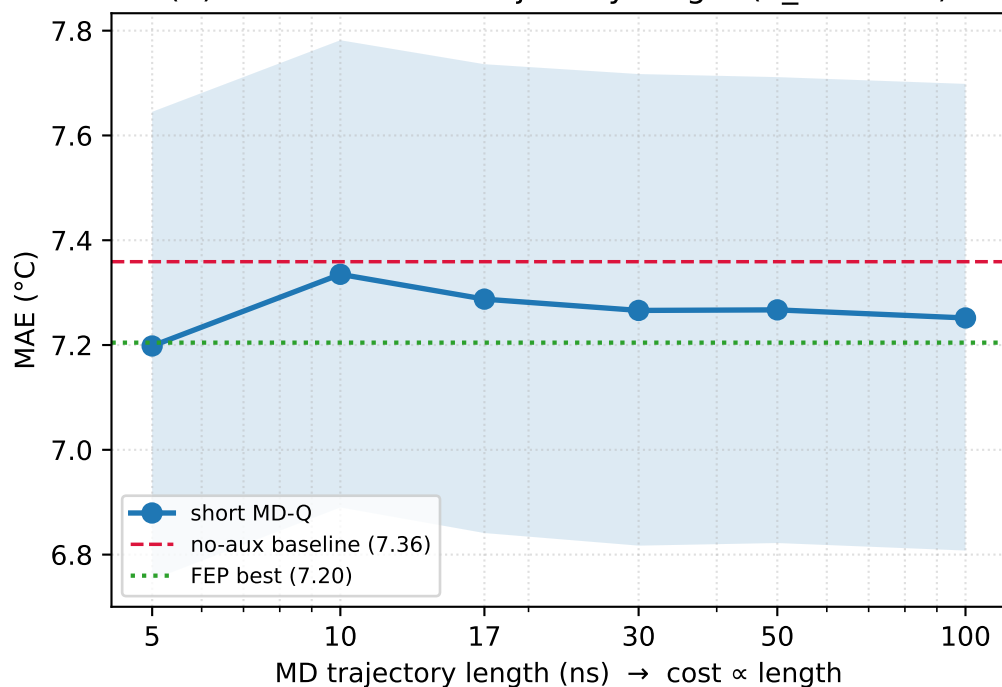
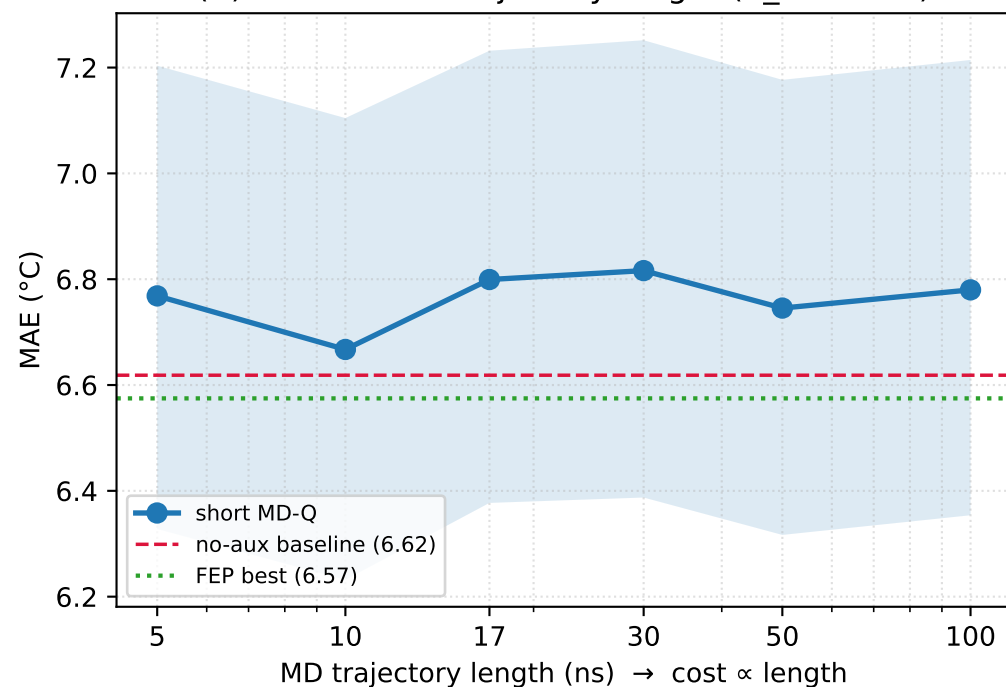


Cost analysis: short (5 ns) MD-Q matches long MD and FEP; longer trajectories add Q range but not accuracy

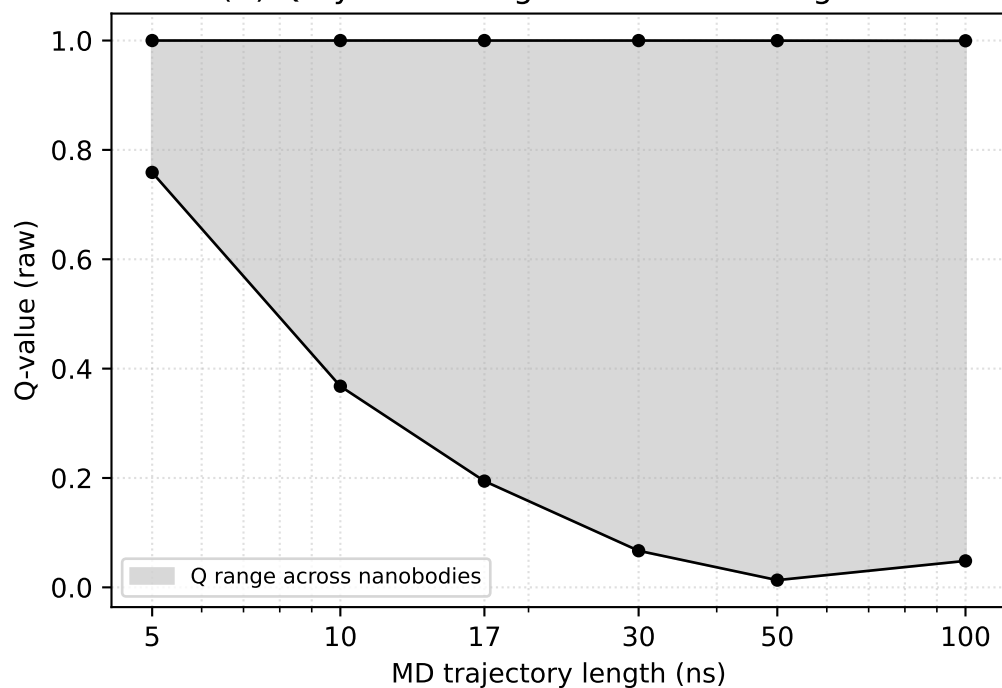
(A) frozen: MAE vs trajectory length (n_md=640)



(B) hot: MAE vs trajectory length (n_md=640)



(C) Q dynamic range widens with length...



(D) ...but accuracy gain stays flat (≈ 0)

